To'g'ri kasrni natural songa bo'lish

Bir bo'lak tarvuzni ikki kishi teng bo'lishsa, har biriga butun tarvuzning qancha qismi tegadi?

1-QISM

Mavzu

ESLAB OLAYLIK

- Sonning qismi ko'paytirish bilan topiladi: $\frac{3}{8}$ ning $\frac{2}{3}$ qismi $= \frac{3}{8} \cdot \frac{2}{3} = \frac{6}{24}$.
- Javobni qisqartiramiz: $\frac{6}{24} = \frac{1}{4}$.

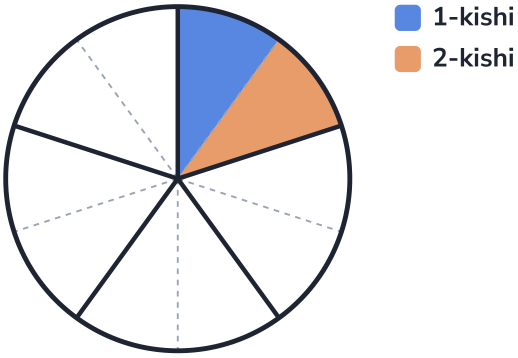
1-QOIDA · BO'LISH VA QISM

«$\frac{3}{4} : 5$» va «$\frac{3}{4}$ ning $\frac{1}{5}$ qismi» bir xil ma'noga ega.

$\frac{3}{4}$ ning $\frac{1}{5}$ qismi $= \frac{3}{4} \cdot \frac{1}{5} = \frac{3}{20}$

1-MISOL · TARVUZ

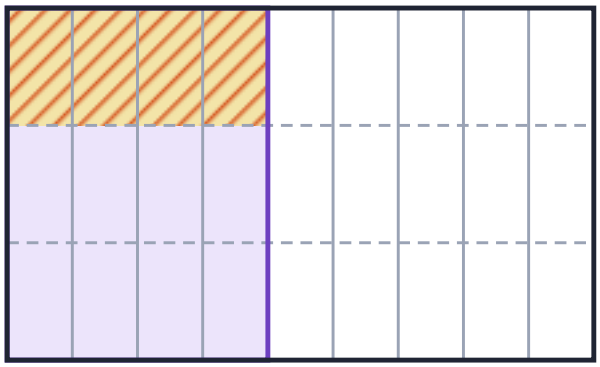
Tarvuzning $\frac{1}{5}$ qismi ikki kishiga teng bo'lindi. Bitta kishining ulushi butun tarvuzning qancha qismi?



1. $\frac{1}{5} : 2$ — bu $\frac{1}{5}$ ning $\frac{1}{2}$ qismi.
2. $\frac{1}{5} \cdot \frac{1}{2} = \frac{1}{10}$ — 10 bo'lakdan bittasi.

2-MISOL · GULZOR

Bog'ning $\frac{4}{9}$ qismi gulzor. Gulzorning uchdan biriga atirgul ekilgan. Atirgul butun bog'ning qancha qismida o'sadi?



bo'yalgan: $\frac{4}{9}$ shtrixlangan: 4 ta katak, bitta butun — 27 ta katak. $\frac{4}{9} \cdot \frac{1}{3} = \frac{4}{27}$

1. $\frac{4}{9}$ ning $\frac{1}{3}$ qismi — bu $\frac{4}{9} : 3$.
2. $\frac{4}{9} \cdot \frac{1}{3} = \frac{4}{27}$ — 27 katakdan 4 tasi.

2-QOIDA · KASRNI NATURAL SONGA BO'LISH

$$\frac{2}{7} : 3 = \frac{2}{7} \cdot \frac{1}{3} = \frac{2}{7 \cdot 3} = \frac{2}{21}$$

- $\frac{2}{7}$ — **o'z holida qoldiring**;
- «:» ni «·» bilan **almashtiring**;
- 3 ning **teskarisini yozing**: $\frac{1}{3}$.

Natural sonning teskarisi — surati 1, maxraji shu son bo'lgan kasr: 3 ning teskarisi $\frac{1}{3}$, 8 ning teskarisi $\frac{1}{8}$.

1-MISOL · QISQARMAYDI

$\frac{5}{8} : 3$ ni toping.

1. $\frac{5}{8} : 3 = \frac{5}{8} \cdot \frac{1}{3}.$
2. $\frac{5}{8 \cdot 3} = \frac{5}{24}.$

2-MISOL · QISQARTIRIB

$\frac{6}{7} : 4$ qancha?

1. $\frac{6}{7} : 4 = \frac{6}{7} \cdot \frac{1}{4}.$
2. Surat va maxrajdagi sonlarni 2 ga bo'lamiz: $6 : 2 = 3$, $4 : 2 = 2$. Hosil bo'ladi: $\frac{3}{7} \cdot \frac{1}{2}.$
3. $\frac{3}{7} \cdot \frac{1}{2} = \frac{3}{14}.$

Avval ko'paytirib, keyin qisqartirsangiz ham shu chiqadi: $\frac{6}{28} = \frac{3}{14}.$



Qarang: surat o'zgarmaydi, maxraj esa songa ko'payadi. $\frac{5}{8} : 3$ da maxraj $8 \cdot 3 = 24$ bo'ldi.

USUL · TENG BO'LISH MASALASI

Teng bo'lishsa, jami miqdorni odamlar (idishlar) soniga bo'lamiz.

1-MISOL · YO'LAK

Bog'dagi $\frac{5}{6}$ km yo'lakni 4 ta usta teng bo'lib asfaltladi. Har bir usta necha kilometr yo'lak asfaltladi?



1. Jami $\frac{5}{6}$ km, ustalar 4 ta.
2. $\frac{5}{6} : 4 = \frac{5}{6} \cdot \frac{1}{4} = \frac{5}{24}$ km.

2-MISOL · QAYMOQ

Qandolatchi $\frac{7}{10}$ l qaymoqni 2 ta tortga teng surtdi. Har bir tortga necha millilitr qaymoq ketdi?

1. $\frac{7}{10} : 2 = \frac{7}{10} \cdot \frac{1}{2} = \frac{7}{20}$ l.
2. 1 l ning $\frac{1}{20}$ qismi $= 1000$ ml ning $\frac{1}{20}$ qismi $= 1000 : 20 = 50$ ml.
3. $\frac{7}{20}$ l $= 7 \cdot 50 = 350$ ml.

2-QISM

Misollar ishlash

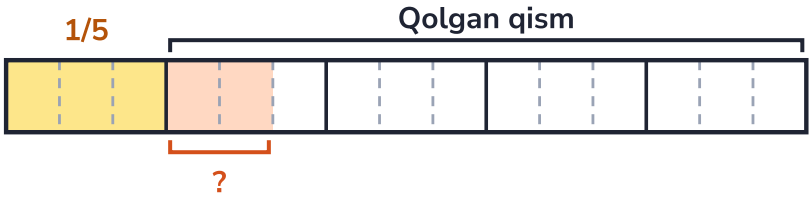
1-MISOL · KATTA SON

$\frac{10}{11} : 15$ bo'linmani toping.

1. $\frac{10}{11} \cdot \frac{1}{15}$.
2. 10 va 15 ni 5 ga bo'lamiz: $10 : 5 = 2, 15 : 5 = 3$. Hosil bo'ladi: $\frac{2}{11} \cdot \frac{1}{3}$.
3. $\frac{2}{11 \cdot 3} = \frac{2}{33}$.

2-MISOL · KITOB

Nodir kitobning $\frac{1}{5}$ qismini o'qidi. Qolgan qismini 6 kunda, har kuni bir xil miqdorda o'qib tugatmoqchi. U har kuni kitobning qancha qismini o'qiydi? 3 kunda-chi?



1. Butun kitob $1 = \frac{5}{5}$.
2. Qolgan qism: $\frac{5}{5} - \frac{1}{5} = \frac{4}{5}$.
3. $\frac{4}{5} : 6 = \frac{4}{5} \cdot \frac{1}{6}$.
4. $4 : 2 = 2$ va $6 : 2 = 3$, hosil bo'ladi $\frac{2}{5} \cdot \frac{1}{3} = \frac{2}{15}$ qism — bir kunda.
5. 3 kunda: $\frac{2}{15} \cdot 3 = \frac{6}{15} = \frac{2}{5}$ qism.

3-MISOL · LENTALAR

Ikki bo'lak lenta bor: $\frac{3}{10}$ m va $\frac{1}{2}$ m. Ular uchma-uch ulanib, 3 ta teng bo'lakka kesildi. Bir bo'lak necha metr?

1. Umumiy maxraj 10 bo'ladi: $\frac{1}{2} = \frac{5}{10}$.
2. Jami: $\frac{3}{10} + \frac{5}{10} = \frac{8}{10} = \frac{4}{5}$ m.
3. $\frac{4}{5} : 3 = \frac{4}{5} \cdot \frac{1}{3} = \frac{4}{15}$ m.

Javob bo'linayotgan kasrdan katta chiqsa, qayerdadir adashgansiz.



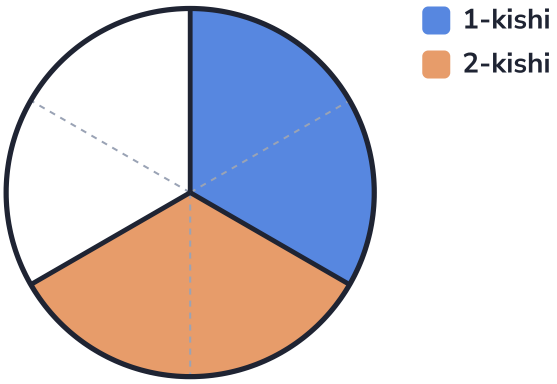
DIQQAT!

1. Bo'luvchi son (3) o'z holida qoldirilmaydi.
   ✗ $\frac{5}{6} : 3 = \frac{5}{6} \cdot 3$ → ✓ $\frac{5}{6} \cdot \frac{1}{3} = \frac{5}{18}$
2. Surat songa ko'paytirilmaydi.
   ✗ $\frac{2}{9} : 4 = \frac{8}{9}$ → ✓ $\frac{2}{9} \cdot \frac{1}{4} = \frac{2}{36} = \frac{1}{18}$
3. Maxraj songa bo'linmaydi.
   ✗ $\frac{7}{10} : 5 = \frac{7}{2}$ → ✓ $\frac{7}{10} \cdot \frac{1}{5} = \frac{7}{50}$
4. ✗ $\frac{8}{9} : 6 = \frac{8}{54} - \frac{8}{54}$ hali qisqarmagan: ✓ $\frac{4}{27}$.
5. $\frac{1}{5}$ l — bu ✗ 5 ml emas, ✓ $1000 : 5 = 200$ **ml**.

3-QISM

Mustahkamlash

1. Doirada qaysi bo'lish chizilgan? Har bir kishiga butunning qancha qismi tegdi?



2. a) «$\frac{5}{7}$ ning $\frac{1}{3}$ qismi» ni bo'lish ko'rinishida yozing va hisoblang. b) «$\frac{1}{6} : 4$» ni «... ning ... qismi» ko'rinishida yozing va hisoblang.

3. Bo'lishni bajaring: a) $\frac{1}{5} : 3$; b) $\frac{2}{7} : 5$; c) $\frac{3}{10} : 2$; d) $\frac{4}{9} : 5$.

4. Javobni qisqarmas kasr ko'rinishida yozing: a) $\frac{6}{11} : 3$; b) $\frac{8}{15} : 4$; c) $\frac{10}{13} : 5$.

5. Qiymatini toping: a) $\frac{3}{4} : 9$; b) $\frac{10}{11} : 4$; c) $\frac{6}{7} : 9$; d) $\frac{9}{16} : 6$.
6. Bo'sh kataklarni to'ldiring: a) $\frac{5}{9} : 2 = \frac{5}{9} \cdot \square = \square$; b) $\frac{3}{8} : \square = \frac{3}{8} \cdot \frac{1}{7} = \square$.
7. **Xatoni toping.** Lola: $\frac{4}{9} : 2 = \frac{8}{9}$. Sardor: $\frac{4}{9} : 2 = \frac{4}{18} = \frac{2}{9}$. Kim to'g'ri hisoblagan? Boshqasi qayerda adashgan?
8. Taqqoslang ($>$, $<$ yoki $=$): a) $\frac{1}{7} : 2$ va $\frac{1}{2} : 7$; b) $\frac{3}{4} : 3$ va $\frac{1}{4}$; c) $\frac{2}{5} : 4$ va $\frac{2}{5} : 5$.
9. Qaysi kasr 2 ga bo'linganda $\frac{1}{18}$ chiqadi: $\frac{1}{9}$, $\frac{1}{36}$ yoki $\frac{2}{36}$?
10. Avval qavs ichini hisoblang: a) $\left(\frac{1}{2} + \frac{1}{4}\right) : 3$; b) $\left(1 - \frac{2}{7}\right) : 5$.
11. $\frac{7}{8}$ ni avval 2 ga, keyin chiqqan natijani 7 ga bo'ling.

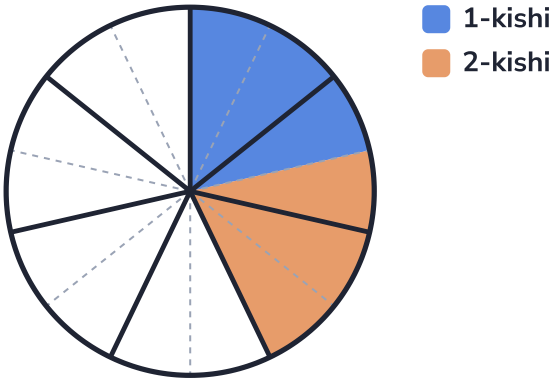
Matnli masalalar

1. Bo'yoqchi $\frac{8}{11}$ kg bo'yoqni 4 ta eshikka teng sarfladi. Bitta eshikka necha kilogramm bo'yoq ketdi?
2. Sayyohlar tog' so'qmog'ining $\frac{5}{7}$ km qismini ikki teng bosqichda bosib o'tishdi. Har bir bosqich necha kilometr?
3. Gul do'konida $\frac{2}{5}$ l o'g'itli suv 8 ta tuvakka teng quyildi. Har bir tuvakka necha millilitr suv quyildi?
4. Maktab hovlisining $\frac{5}{8}$ qismini sport maydoni egallaydi. Uni tozalash 10 ta sinfga teng bo'lib berildi. Har bir sinfga hovlining qancha qismi tegdi?
5. Polizning $\frac{6}{7}$ qismiga 9 ta bir xil egat olindi; ulardan 3 tasiga qovun ekildi. Qovun polizning qancha qismiga ekildi? *(Avval bitta egatni toping, keyin 3 ga ko'paytiring.)*

4-QISM

Mustaqil ish

1. Bo'linmalarni toping: a) $\frac{2}{3} : 7$; b) $\frac{9}{14} : 6$; c) $\frac{12}{13} : 8$.
2. Rasmga qarab bo'lishni yozing va natijani toping.



3. $\left(\frac{9}{10} - \frac{1}{2}\right) : 2$ ifodaning qiymatini toping.
4. Rassom $\frac{3}{4}$ l oq bo'yoqni 10 ta kichik bankaga teng quydi. Har bir bankada necha millilitr bo'yoq bor? *(Yordam: 1 l = 1000 ml.)*
5. Qaysi ulush katta: $\frac{3}{8}$ l kompotni 3 kishiga teng bo'lgandami yoki $\frac{5}{6}$ l kompotni 5 kishiga teng bo'lgandami? Qanchaga katta? *(Yordam: ikki javobni maxraji 24 bo'lgan kasrga keltiring.)*
6. Bo'sh katakka qaysi natural son yoziladi: $\frac{3}{7} : \square = \frac{3}{35}$?

5-QISM

Javoblar

Mustahkamlash

1-masala

$\frac{2}{3} : 2 = \frac{2}{3} \cdot \frac{1}{2} = \frac{2}{6} = \frac{1}{3}.$

2-masala

a) $\frac{5}{7} : 3 = \frac{5}{7} \cdot \frac{1}{3} = \frac{5}{21}$. b) $\frac{1}{6}$ ning $\frac{1}{4}$ qismi $= \frac{1}{6} \cdot \frac{1}{4} = \frac{1}{24}$.

3-masala

a) $\frac{1}{15}$; b) $\frac{2}{35}$; c) $\frac{3}{20}$; d) $\frac{4}{45}$.

4-masala

a) $\frac{6}{33} = \frac{2}{11}$; b) $\frac{8}{60} = \frac{2}{15}$; c) $\frac{10}{65} = \frac{2}{13}$.

5-masala

a) $3 : 3 = 1$ va $9 : 3 = 3$, hosil bo'ladi $\frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12}$; b) $\frac{5}{11} \cdot \frac{1}{2} = \frac{5}{22}$; c) $\frac{2}{7} \cdot \frac{1}{3} = \frac{2}{21}$; d) $\frac{3}{16} \cdot \frac{1}{2} = \frac{3}{32}$.

6-masala

a) $\frac{5}{9} \cdot \frac{1}{2} = \frac{5}{18}$; b) bo'sh katakda 7, natija $\frac{3}{56}$.

7-masala

Sardor to'g'ri: Lola suratni 2 ga ko'paytirgan.

8-masala

a) $\frac{1}{14} = \frac{1}{14}$; b) $\frac{1}{4} = \frac{1}{4}$; c) $\frac{1}{10} > \frac{2}{25}$, chunki $\frac{5}{50} > \frac{4}{50}$.

9-masala

$\frac{1}{9} : 2 = \frac{1}{18}$. Qolganlari: $\frac{1}{36} : 2 = \frac{1}{72}$, $\frac{2}{36} : 2 = \frac{1}{36}$.

10-masala

a) $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$; $\frac{3}{4} : 3 = \frac{1}{4}$. b) $\frac{7}{7} - \frac{2}{7} = \frac{5}{7}$; $\frac{5}{7} : 5 = \frac{5}{35} = \frac{1}{7}$.

11-masala

$\frac{7}{8} : 2 = \frac{7}{16}$; $\frac{7}{16} : 7 = \frac{7}{112} = \frac{1}{16}$.

Matnli masalalar

1-masala

$\frac{8}{11} : 4 = \frac{8}{44} = \frac{2}{11}$ kg.

2-masala

$\frac{5}{7} : 2 = \frac{5}{14}$ km.

3-masala

$\frac{2}{5} : 8 = \frac{2}{40} = \frac{1}{20}$ l; $1000 : 20 = 50$ ml.

4-masala

$\frac{5}{8} : 10 = \frac{5}{80} = \frac{1}{16}$ qism.

5-masala

Bitta egat: $\frac{6}{7} : 9 = \frac{6}{63} = \frac{2}{21}$. Uchta egat: $\frac{2}{21} \cdot 3 = \frac{6}{21} = \frac{2}{7}$ qism.

Mustaqil ish

1-masala

a) $\frac{2}{21}$; b) $\frac{9}{84} = \frac{3}{28}$; c) $\frac{12}{104} = \frac{3}{26}$.

2-masala

$\frac{3}{7} : 2 = \frac{3}{7} \cdot \frac{1}{2} = \frac{3}{14}.$

3-masala

$\frac{9}{10} - \frac{5}{10} = \frac{4}{10} = \frac{2}{5}; \frac{2}{5} : 2 = \frac{2}{10} = \frac{1}{5}.$

4-masala

$\frac{3}{4} : 10 = \frac{3}{40}$ l. $\frac{1}{40}$ l $= 1000 : 40 = 25$ ml, $\frac{3}{40}$ l $= 3 \cdot 25 = 75$ ml.

5-masala

$\frac{3}{8} : 3 = \frac{1}{8} = \frac{3}{24}$ l; $\frac{5}{6} : 5 = \frac{1}{6} = \frac{4}{24}$ l. Ikkinchi ulush $\frac{4}{24} - \frac{3}{24} = \frac{1}{24}$ l ga katta.

6-masala

$\frac{3}{7} : 5 = \frac{3}{7 \cdot 5} = \frac{3}{35}$. Bo'sh katakda 5.